

LE NGUYEN GIA HUNG

AI Research Engineer · Time-Series & Multimodal XAI

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EDUCATION

FPT University

Bachelor of Engineering in Artificial Intelligence

Ho Chi Minh City, Vietnam

Expected Dec 2027 GPA: 8.62/10.0

- **Honors:** Certificate of Merit: Honorable Student of Trimester (Semesters 3, 4, 5, 6).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence, Deep Learning, Statistical Analysis.

PUBLICATIONS & MANUSCRIPTS

Co-Authors, **Gia-Hung Nguyen Le** (Second Author) *et al.*, “Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia.”

Manuscript under review.

Gia-Hung Nguyen Le (First & Corresponding Author) *et al.*, “Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring.”

Under Review at the Australasian Joint Conference on Artificial Intelligence (AJCAI 2026).

Co-Authors, **Gia-Hung Nguyen Le** (Second Author) *et al.*, “Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution.”

Under Review at the 33rd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2027).

Gia-Hung Nguyen Le (First & Corresponding Author) *et al.*, “Proactive Air Quality Forecasting and Health Alert System for Melbourne.”

Australasian Joint Conference on Artificial Intelligence (AJCAI 2025). Springer, Lecture Notes in Artificial Intelligence (LNAI). Q2 (SJR 2024).

DOI: 10.1007/978-981-95-4969-6.34

RESEARCH & TECHNICAL EXPERIENCE

Health-Centred Air Quality Alert System (NZ & AU)

Second Author · Manuscript under review

Academic Research

Jan 2026 – Present

- Re-framed air quality alert evaluation from raw concentration accuracy to health-centred exposure coverage and intra-day warning timing across New Zealand and Australia.
- Discovered that regulatory daily-mean triggers overstate health protection by 7.7–18.6 points across 18 configurations, as 42% of exposure occurs on sub-threshold daily mean days.
- Implemented a weather-forecast panel model achieving **MAE of 3.19 $\mu\text{g}/\text{m}^3$** and **R^2 of 0.54** 1-day ahead (outperforming persistence by 26%, gradient boosting by 12%, and sequence baselines by 8%) holding lead up to 3 days.
- Analyzed cross-Tasman smoke dynamics across 10 years of Canterbury (NZ) hourly data and 4 years of Melbourne (AU) network data, contrasting local winter peaks against transported smoke.

Cyanobacterial Bloom Forecasting under Sparse Monitoring

First Author & Corresponding Author · AJCAI 2026 Under Review

Academic Research

Jan 2026 – Present

- Stress-tested multi-source time-series forecasting across 7 New Zealand lakes using a causal weekly benchmark and untouched 2023–2024 holdout ($n = 170$).
- Achieved a weighted **F1 of 0.791** for 7-day horizon forecasting; evaluated horizon-dependent fusion trade-offs and alert burden calibration.
- Developed an executable audit protocol coupling feature availability, naive baselines, incident eligibility, and paired inference contracts.

Trustworthy Multimodal Anomaly Detection in Time Series

Second Author · ACM SIGKDD 2027 Under Review

Academic Research

Jan 2026 – Present

- Developed the **Context-Conditioned Detect-Localize-Explain (C-DLE)** framework for multimodal time-series anomaly detection.
- Implemented **split-conformal calibration** on fused numerical and textual scores, delivering distribution-free control of false alarm rates.
- Designed a cross-modal counterfactual sufficiency rule achieving **94.3% attribution accuracy** against known ground truth.

- Audited temporal text leakage across 9 domains and optimized calibration pools into coresets at **10% size** without skill loss.

Proactive Air Quality Forecasting System (SmokeNet)

Academic Research

First Author & Corresponding Author · AJCAI 2025 Published (LNAI Q2)

Jun 2025 – Nov 2025

- Designed and implemented **SmokeNet**, a Transformer-based architecture for PM_{2.5} time-series forecasting.
- Achieved **MAE 0.7470** and R^2 **0.9545** (35.9% error reduction vs. XGBoost); outperformed baselines by **57.7%** during severe smoke backtests.
- Engineered end-to-end pipeline processing 4+ years of data with GPU cross-validation and integrated Explainable AI (SHAP, Integrated Gradients).

MERR-GAT: Multimodal Emotion Recognition via Graph Attention

Academic Research

Second Author & Corresponding Author · Targeting Elsevier Information Fusion (Q1)

Oct 2025 – Present

- Developing **MERR-GAT**, a novel hybrid framework fusing **Wav2Vec 2.0** (audio) and **RoBERTa** (text) via Graph Attention Networks (GATv2).
- Designing a dynamic conversational graph where nodes represent utterances and edges capture temporal and speaker interactions.
- Implementing an intrinsic XAI mechanism where learned attention weights provide granular insights into the model's decision-making.
- Benchmarking on **IEMOCAP**, **RAVDESS**, and **MELD** with a dynamic classification head supporting 4, 5, or 6 emotion classes.

End-to-End Vietnamese Speech Recognition System

Academic Research

Co-Author · Speech Corpus Benchmark

Sep 2025 – Present

- Developing a parameter-efficient (5.4M) end-to-end ASR model using a **4-layer CNN frontend** and **3-layer Residual BiLSTM encoder**.
- Curated and trained on a massive **745-hour Vietnamese speech corpus**, optimizing with Connectionist Temporal Classification (CTC) loss.
- Achieved a **Word Error Rate (WER) of 33.09%** and **Character Error Rate (CER) of 15.28%** on diverse test sets; benchmarked against **PhoWhisper**.

LEADERSHIP

SpeedyLabX - Student Research Group

FPT University

Founder & Student Lead

2025 – Present

- Founded a student research group of 10 undergraduates; led weekly technical workshops and reading groups.
- Coordinated submission workflow (milestones, reviews, camera-ready) for the group's first AJCAI-accepted paper.

CERTIFICATIONS

- **Data Science Fundamentals with Python and SQL** – IBM
- **Statistics for Data Science with Python** – IBM
- **Introduction to Containers w/ Docker, Kubernetes & OpenShift** – IBM

SKILLS

- **Languages:** Python, SQL (PostgreSQL).
- **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers, PyTorch Geometric (PyG / GATv2), TensorFlow/Keras, Scikit-learn, OpenCV.
- **Explainable AI (XAI) & Evaluation:** SHAP, Integrated Gradients, Chronological CV, CTC Loss, Weights & Biases (W&B).
- **Data & MLOps/Tools:** Pandas, NumPy, Matplotlib, Git, Docker, Linux, Streamlit.
- **Technical writing:** Blog posts and technical documentation (portfolio: hei-portfolio.vercel.app).